Abhinav Kumar

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Summary

Scientific Software Engineer with a PhD in Physics, experienced in building Python and C/C++/Fortran codes for machine learning, numerical simulations, optimization, high-throughput HPC workflows, and large-scale data analysis. Trained in CUDA C/C++ and multi-GPU programming.

Skills

Programming	Python (Pandas, NumPy, SciPy, Matplotlib), C++, C, Fortran, Git (version control), Matlab, IDL
HPC	Bash scripting (Linux/Unix), SLURM job arrays, distributed/parallel job execution, high-throughput workflows
Scientific	Statistical data analysis, mathematical modelling, numerical simulations, parameter estimation, optimization techniques, stochastic systems, interpolation, Fourier transformations, linear algebra, monte carlo methods
Machine Learning	Neural networks, supervised learning, model training and evaluation, hardware-aware ML, NumPy-based ML implementation
GPU Computing	CUDA C/C++, GPU kernels, memory management, streams, Nsight Systems, multi-GPU programming, CUDA-aware MPI, and NVSHMEM

Machine Learning

Neuromorphic Computing Using Experimental Synaptic-Device Data *2026*

- Built a Python pipeline that converts experimental synaptic-device measurements into fitted mathematical models and integrates them into an MLP for MNIST classification.
- Implemented neural-network training from scratch in Python, including forward and backpropagation, Adam optimization, and ReLU activation.

Neural Network Visualizer | Live Demo | GitHub *2026*

- Built and deployed an interactive MNIST visualizer with real browser inference and training evolution.

Professional Experience

Postdoctoral Researcher *Germany*

Physics of Life, Technische Universität Dresden *Aug 2025 - Present*

- Simulated stochastic systems and applied mathematical modeling through simulations on high-performance computing (HPC) clusters.
- Extended the open source C++ molecular dynamics simulation package ESPResSo by implementing custom interactions.
- Transformed simulation outputs into interpretable visual and quantitative insights in 2D and 3D, using Python.

Graduate Research Assistant *Merced, CA, USA*

Department of Physics, University of California, Merced *Aug 2019 - Aug 2025*

- Developed simulation codes in C, implementing numerical optimization algorithms.
- Automated execution of thousands of simulations on high-performance computing (HPC) clusters using bash script.
- Built Python workflows for large-scale data analysis, statistical modeling, and visualizations.

Software Engineer *Mumbai, India*

EnerNOC *Jan 2016 - July 2016*

- **Awards**: Employee of the Month (April, 2016)
- Maintained and extended legacy Fortran code used in production energy analytics workflows.

Education

University of California, Merced *California, USA*

PhD in Physics *2019-25*

Sikkim Manipal University *Sikkim, India*

Master of Science in Physics *2016-18*

University of Kansas *Kansas, USA*

Bachelor of Science in Physics, Astronomy *2010-14*

Publications

- Cooperative effect of local active stresses on the macroscopic contractility of elastic fiber networks, **Abhinav Kumar**, David A. Quint, Kinjal Dasbiswas, **Soft Matter, 2026,22, 3790-3805** DOI: `https://doi.org/10.1039/D5SM00803D`
- Range and strength of mechanical interactions of force dipoles in elastic fiber networks, **Abhinav Kumar**, David A. Quint, Kinjal Dasbiswas, **Soft Matter, 2023, 19, 5805-5823** DOI: `https://doi.org/10.1039/D3SM00381G`